An Erdős–Turán Law for Comparable-Rate Luce Permutations

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Abstract

Let σ_n be the Luce permutation induced by independent exponential clocks whose rates, after a common rescaling, satisfy $1 \le \theta_{n,i} \le \kappa$ for one fixed $\kappa < \infty$. We prove

$$\frac{\log \operatorname{ord}(\sigma_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \Rightarrow N(0,1).$$

Thus the classical Erdős–Turán law persists for arbitrary comparable-rate triangular arrays; no limiting rate profile is required. The proof treats the additive proxy $\sum_r C_{n,r} \log r$ by a weighted clock-swap characteristic-function argument. A quantitative finite-endpoint comparison is used through a theorem-level interface. The arithmetic difference between the product and the least common multiple of the cycle lengths is controlled by an exact rate-oriented three-clock switch, whose orientation leaves only one positive coefficient in the Radon–Nikodym logarithm and yields a uniform two-cycle factorial-moment bound.

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1 Introduction

For a permutation $\pi \in S_n$, its order is the least positive integer k for which π^k is the identity, equivalently

$$\operatorname{ord}(\pi) = \operatorname{lcm}\{\text{cycle lengths of } \pi\}.$$

For a uniformly chosen permutation, Erdős and Turán proved that the logarithm of the order is asymptotically Gaussian on the $(\log n)^{3/2}$ scale [5]. Sharp rates and a broader probabilistic theory were subsequently developed for the uniform and Ewens settings and for logarithmic combinatorial structures; see, among others, [2, 1, 6, 12].

The Luce distribution is generated by ranking independent exponential clocks with prescribed rates. It is the stationary distribution of the Tsetlin library and is typically nonexchangeable and spatially inhomogeneous [11, 4]. Recent work developed the permuton, local, and multiscale cycle theories of this model [3, 9, 10]. The permutation order is a particularly sensitive remaining statistic: it depends on divisibility relations among cycle lengths across fixed, mesoscopic, and macroscopic scales, and is not determined by a permuton limit or by macroscopic Poisson–Dirichlet convergence alone.

Our result shows that the classical order law is nevertheless universal throughout the comparable-rate class.

Theorem 1.1 (Main theorem)**.** *Let σ_n be a Luce permutation with positive rates $\theta_{n,1}, \dots, \theta_{n,n}$ satisfying*

$$\sup_n \frac{\max_i \theta_{n,i}}{\min_i \theta_{n,i}} < \infty.$$

Then

$$\frac{\log \operatorname{ord}(\sigma_n) - \frac{1}{2}(\log n)^2}{\sqrt{\frac{1}{3}(\log n)^3}} \Rightarrow N(0,1).$$

The theorem is stronger in hypothesis scope than a sampled-profile result: the rates may form an arbitrary triangular array, and neither convergence nor regularity of a macroscopic profile is assumed.

Proof architecture

After a harmless common rescaling, take $1 \le \theta_{n,i} \le \kappa$. The argument has four parts.

(i) The product of the cycle lengths has logarithm

$$B_n = \sum_{r=1}^{n} C_{n,r} \log r.$$

A weighted clock-swap identity and a quantitative finite-endpoint comparison yield the same characteristic exponent as an independent Poisson cycle process, up to an error negligible on the $(\log n)^{3/2}$ scale.

(ii) The quantitative finite-endpoint theory is imported only through the precise Input Theorem 3.1. All subsequent summations and optimizations are carried out in this paper.

(iii) A new rate-oriented three-clock change of measure controls two cycle counts simultaneously. The cyclic orientation is chosen by decreasing rate, which makes the Radon–Nikodym logarithm have exactly one positive clock coefficient and gives a uniform $C_\kappa/(rs)$ factorial-moment bound.

(iv) The von Mangoldt decomposition converts the difference between the cycle product and the cycle least common multiple into prime-power overlap counts. The one- and two-cycle estimates imply an $O_\kappa(\log n \log\log n)$ bound in mean, which is negligible at the Erdős–Turán scale.

Dependencies and quantitative refinements

The only imported probabilistic inputs are the marked one-switch estimates in Input Theorem 2.1 and the quantitative finite-endpoint interface in Input Theorem 3.1, both from the companion switching manuscript [10]. The weighted characteristic-function argument, the rate-oriented three-clock identity, and the arithmetic product–LCM comparison are proved here. A separate quantitative supplement [8] proves the sharp $O_\kappa((\log n)^{-1/2})$ Kolmogorov rate under exact-mean centering and a first Edgeworth expansion. That argument uses a stronger linear-depth form of the finite-endpoint comparison and fixed-order multi-clock estimates; neither strengthening is needed for the qualitative theorem proved here.

2 Model, switching inputs, and cutoffs

Let

$$X_{n,i} \sim \mathrm{Exp}(\theta_{n,i}), \qquad 1 \le i \le n,$$

be independent, with

$$1 \le \theta_{n,i} \le \kappa. \tag{1}$$

Let $\sigma_n(i)$ be the rank of $X_{n,i}$ among the n clocks. Write $C_{n,r}$ for the number of r-cycles of σ_n and $L_n(i)$ for the length of the cycle containing i.

For distinct labels i, j, put

$$R_{ij} = \exp\{-(\theta_i - \theta_j)(X_j - X_i)\}, \qquad R_{ii} = 1,$$

and define the switching intensity

$$S_{n,r} = \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}.$$

We import the following estimates from the clock-swap and finite-endpoint mixing theory in [10].

Input theorem 2.1 (Marked one-switch identity, moments, and tails)**.** *For every $1 \le r < n$ and every bounded complex-valued function $\Phi : S_n \to \mathbb{C}$,*

$$r(n-r)\mathbb{E}\big[C_{n,r}\Phi(\sigma_n)\big] = \mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\Phi\big(\sigma_n \circ (i\,\sigma_n^r(i))\big). \tag{2}$$

Taking $\Phi \equiv 1$ gives

$$r(n-r)\mathbb{E}C_{n,r} = \mathbb{E}S_{n,r}. \tag{3}$$

Set

$$\lambda_\kappa = \begin{cases} \infty, & \kappa = 1, \\ \dfrac{\kappa}{\kappa - 1}, & \kappa > 1. \end{cases}$$

For every $1 < p < \lambda_\kappa$,

$$\sup_{n,r} \frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R^p_{i,\sigma_n^r(i)} < \infty. \tag{4}$$

If $\kappa > 1$, then, uniformly for $A \geq 1$,

$$\sup_{n,r} \frac{1}{n} \mathbb{E} \sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)} \mathbf{1}_{\{R_{i,\sigma_n^r(i)}>A\}} \leq C_\kappa A^{-1/(\kappa-1)} \tag{5}$$

$$\sup_n \max_{i,j} \mathbb{E}\left[R_{ij} \mathbf{1}_{\{R_{ij}>A\}}\right] \leq C_\kappa A^{-1/(\kappa-1)}. \tag{6}$$

Moreover, $\mathbb{E}R_{ij} = 1$ for every i, j. If $\kappa = 1$, all the factors R_{ij} are identically one.

In particular, $\mathbb{E}S_{n,r} \leq C_\kappa n$ uniformly in n, r.

Lemma 2.2 (Crude cycle, root, and logarithmic-tail bounds)**.** *Uniformly in $1 \leq r < n$,*

$$\mathbb{E}C_{n,r} \leq \frac{C_\kappa n}{r(n-r)}. \tag{7}$$

If I_n is uniform on $[n]$, independently of the clocks, then, for $a \leq n/2$,

$$\mathbb{P}\{L_n(I_n) \leq a\} \leq C_\kappa a/n. \tag{8}$$

Consequently,

$$\mathbb{E}\sum_{r\leq b} (\log r) C_{n,r} = O_\kappa((\log b)^2), \tag{9}$$

and, for $U \leq n/2$,

$$\mathbb{E}\sum_{r>U} (\log r) C_{n,r} \leq C_\kappa \log n\{1 + \log(n/U)\}. \tag{10}$$

Proof. Equation (7) follows from (3) and $\mathbb{E}S_{n,r} \leq C_\kappa n$. Since

$$\mathbb{P}\{L_n(I_n) = r\} = \frac{r\mathbb{E}C_{n,r}}{n},$$

for $a \leq n/2$ we have

$$\mathbb{P}\{L_n(I_n) \leq a\} \leq C_\kappa \sum_{r\leq a} \frac{1}{n-r} \leq C'_\kappa \frac{a}{n}.$$

For $r \leq n/2$, equation (7) gives $\mathbb{E}C_{n,r} \leq 2C_\kappa/r$. Summing $(\log r)/r$ proves (9). For $U < r \leq n/2$, the same estimate gives

$$\mathbb{E}\sum_{U<r\leq n/2} (\log r) C_{n,r} \leq C_\kappa \log n \sum_{U<r\leq n/2} \frac{1}{r}.$$

There is deterministically at most one cycle of length greater than $n/2$, which contributes at most $\log n$. This proves (10). □

Fix once and for all a number p satisfying

$$1 < p < \lambda_\kappa$$

(with any $p > 1$ when $\kappa = 1$), and set

$$\beta = 1 - \frac{1}{p} > 0.$$

Choose a fixed constant

$$A_0 > \max\left\{2, \frac{1}{2\beta}\right\}, \tag{11}$$

and put

$$b_n = \left\lceil (\log n)^3 \right\rceil, \qquad U_n = \left\lfloor \frac{n}{(\log n)^{A_0}} \right\rfloor. \tag{12}$$

For all sufficiently large n,

$$2U_n \le n/2. \tag{13}$$

3 Quantitative switching on dyadic blocks

Put

$$\Delta_{n,r} = \mathbb{E}\left|\frac{S_{n,r}}{n} - 1\right|.$$

Fix an integer a with $b_n \le a \le U_n$. In the two-root endpoint argument take lower length a, upper length $2a$, and hence

$$\delta = \frac{4a}{n}, \qquad \ell = \log\frac{1}{\delta}, \qquad h = \delta^{1/4}, \qquad \rho = \delta^{1/(4\kappa)}, \qquad M = n\delta^{1/2}, \tag{14}$$

and

$$m = \left\lfloor \min\left\{\frac{a}{3}, \sqrt{\ell}\right\}\right\rfloor. \tag{15}$$

These are the parameters of the finite-endpoint construction with two independent roots and maximal exposed length $2a$.

For $A \ge 1$, define

$$F_A(i,j) = R_{ij} \wedge A, \qquad S_{n,r}^{(A)} = \sum_{i:L_n(i)>r} F_A(i, \sigma_n^r(i)),$$

and

$$D_{n,A} = \frac{1}{n^2}\sum_{i,j=1}^{n} F_A(i,j), \qquad \mu_{n,A} = \mathbb{E}D_{n,A}.$$

The following is the complete interface from the finite-endpoint theory used in this paper. It is the quantitative form of the one- and two-root endpoint comparison proved in Sections 6–8 of [10].

Input theorem 3.1 (Quantitative finite-endpoint comparison)**.** *There are constants $c, C, \alpha_0 > 0$, depending only on κ, such that, uniformly for $b_n \le a \le U_n$, $a \le r < 2a$, and $A \ge 1$,*

$$\left|\mathbb{E}\frac{S_{n,r}^{(A)}}{n} - \mu_{n,A}\right| \le C_\kappa A \Gamma_{n,a}, \tag{16}$$

$$\left|\mathbb{E}\left(\frac{S_{n,r}^{(A)}}{n}\right)^2 - \mathbb{E}D_{n,A}^2\right| \le C_\kappa A^2 \Gamma_{n,a}, \tag{17}$$

where

$$\Gamma_{n,a} \le C\Bigg[(1-\alpha_0)^{m-1} + m\delta^{1/4} + \frac{m^2}{n\delta^{1/2}} + m\delta^{1/(4\kappa)} + \frac{m\log n}{n\delta^{1/2}} + \delta^{1/(4\kappa)} + \delta^{1/2}\ell + \delta + g_{n,a}\Bigg], \tag{18}$$

with

$$g_{n,a} \le C\left[\delta^{-1/4}\ell\, e^{-cn\delta^{1/2}} + n^2\delta^{-1/4}e^{-cn\delta^{3/4}} + n^{-2}\right]. \tag{19}$$

The constants are independent of the truncation level A.

Remark 3.2 (Use of the input theorem). Only the bounds (16)–(19) are used below. In particular, the clock-block construction, conditional permanental matching, reset chain, and adaptive exposure bookkeeping from the companion paper do not enter the present proof except through this stated interface. The companion endpoint theorem admits a free terminal-depth parameter. On the present dyadic range, this paper uses the square-root-depth specialization $m \asymp \sqrt{\ell}$, which suffices for the qualitative characteristic-function argument, whereas the quantitative supplement [8] uses $m \asymp \ell$ to obtain polynomial product–Poisson accuracy.

Proposition 3.3 (One-block quantitative switching)**.** *There is $C_\kappa < \infty$ such that, uniformly for $b_n \le a \le U_n$ and $a \le r < 2a$,*

$$\Delta_{n,r} \le C_\kappa(\Gamma_{n,a} + n^{-1})^{1/(2\kappa)}. \tag{20}$$

Proof. Replacing one clock X_k changes only the terms of $D_{n,A}$ with $i = k$ or $j = k$, at most $2n$ terms, each lying in $[0, A]$. Hence $|D_{n,A} - D_{n,A}^{(k)}| \le 2A/n$, and the Efron–Stein inequality gives

$$\operatorname{Var}(D_{n,A}) \le \frac{4A^2}{n}. \tag{21}$$

Put $X_{n,r}^{(A)} = S_{n,r}^{(A)}/n$. Since $0 \le \mu_{n,A} \le 1$,

$$\begin{aligned}\mathbb{E}|X_{n,r}^{(A)} - \mu_{n,A}|^2 = \big(\mathbb{E}(X_{n,r}^{(A)})^2 - \mathbb{E}D_{n,A}^2\big) + \operatorname{Var}(D_{n,A})\\ - 2\mu_{n,A}\big(\mathbb{E}X_{n,r}^{(A)} - \mu_{n,A}\big).\end{aligned}$$

Equations (16), (17), and (21) therefore imply

$$\mathbb{E}|X_{n,r}^{(A)} - \mu_{n,A}|^2 \le C_\kappa A^2(\Gamma_{n,a} + n^{-1}). \tag{22}$$

Suppose first that $\kappa > 1$. Equations (5) and (6), together with $\mathbb{E}R_{ij} = 1$, give

$$\mathbb{E}\frac{S_{n,r} - S_{n,r}^{(A)}}{n} + |1 - \mu_{n,A}| \le C_\kappa A^{-1/(\kappa-1)}.$$

By (22),

$$\Delta_{n,r} \le C_\kappa\left[A(\Gamma_{n,a} + n^{-1})^{1/2} + A^{-1/(\kappa-1)}\right]. \tag{23}$$

For $\Gamma_{n,a} + n^{-1} \le 1$, choose

$$A = (\Gamma_{n,a} + n^{-1})^{-(\kappa-1)/(2\kappa)};$$

when the quantity exceeds one, the claimed bound follows after enlarging the constant. This proves (20). If $\kappa = 1$, all swap factors equal one, no truncation is needed, and (22) with $A = 1$ gives the same conclusion with exponent $1/2 = 1/(2\kappa)$. □

Corollary 3.4 (Logarithmically weighted dyadic summability)**.** *Let $a_j = 2^j b_n$. Then*

$$\sum_{j:\, a_j \le U_n} \log(2a_j) \sup_{a_j \le r < 2a_j} \Delta_{n,r} = o((\log n)^{3/2}). \tag{24}$$

Proof. Set

$$\eta = \frac{1}{2\kappa}, \qquad \delta_j = \frac{4a_j}{n}, \qquad L_j = \log \frac{1}{\delta_j}.$$

Since $a_j \geq b_n = (\log n)^3$ and $L_j \leq \log n$, the minimum in (15) is attained by $\sqrt{L_j}$ for all sufficiently large n, uniformly in j. Thus $m_j \asymp \sqrt{L_j}$. Moreover,

$$L_j \geq A_0 \log\log n - O(1).$$

Since $0 < \eta \leq 1$, the map $t \mapsto t^\eta$ is subadditive on $[0, \infty)$. We may therefore sum the terms in (18) separately after applying Proposition 3.3.

For the reset-mixing term,

$$\begin{aligned}\sum_j \log(2a_j) e^{-c\eta m_j} &\leq C \log n \sum_{L_j \geq A_0 \log\log n - O(1)} e^{-c'\sqrt{L_j}} \\ &\leq C \log n \,(1 + \sqrt{\log\log n}) e^{-c''\sqrt{\log\log n}} = o((\log n)^{3/2}).\end{aligned}$$

After taking the η-power, each term involving a positive power of δ_j is bounded by

$$C L_j^d e^{-\gamma L_j}$$

for fixed $d \geq 0$ and $\gamma > 0$. Since the L_j form an arithmetic lattice with fixed spacing,

$$\sum_j L_j^d e^{-\gamma L_j} \leq C(\log\log n)^d (\log n)^{-\gamma A_0}.$$

Multiplication by the weight $\log(2a_j) \leq \log n$ is still $o((\log n)^{3/2})$. This treats $m\delta^{1/4}$, $m\delta^{1/(4\kappa)}$, $\delta^{1/(4\kappa)}$, $\delta^{1/2}\ell$, and δ.

For the terms containing $M^{-1} = (n\delta_j^{1/2})^{-1}$, use $m_j \leq \sqrt{\log n}$ and $a_j \geq (\log n)^3$ to obtain

$$\frac{m_j^2 + m_j \log n}{n\delta_j^{1/2}} \leq C \frac{(\log n)^{3/2}}{\sqrt{n a_j}}.$$

Since $a_j \geq (\log n)^3$, the apparent factor $(\log n)^{3/2}$ is cancelled by $\sqrt{a_j}$. After taking the η-power, multiplying by a logarithmic weight, and summing over $O(\log n)$ blocks, the contribution is therefore at most

$$C(\log n)^2 \left(\frac{1}{\sqrt{n}}\right)^\eta = o(1).$$

The first two terms of (19) are superpolynomially small uniformly for $a_j \geq (\log n)^3$. The n^{-2} term and the additional n^{-1} in (20) contribute at most $C(\log n)^2 n^{-\eta} = o(1)$. Combining the estimates proves (24). □

4 A central limit theorem for the product of the cycle lengths

Define

$$B_n = \sum_{r=1}^{n} (\log r) C_{n,r} = \log \prod_{C \in \mathrm{Cyc}(\sigma_n)} |C|, \tag{25}$$

and its truncated version

$$B_n^\circ = \sum_{b_n < r \leq U_n} (\log r) C_{n,r}. \tag{26}$$

Lemma 4.1 (Truncation of the additive proxy)**.** *One has*

$$\mathbb{E}(B_n - B_n^\circ) = O_\kappa(\log n \log\log n). \tag{27}$$

In particular,

$$B_n - B_n^\circ = o_{L^1}((\log n)^{3/2}).$$

Proof. By (9), the contribution from $r \le b_n$ is $O((\log b_n)^2) = O((\log\log n)^2)$. By (10), the contribution from $r > U_n$ is

$$O\big(\log n\{1 + \log(n/U_n)\}\big) = O(\log n \log\log n).$$

This proves (27). □

Let

$$Z_n(s) = \mathbb{E}e^{sB_n^\circ}, \qquad a_{n,r} = \frac{n}{r(n-r)}, \qquad w_r = \log r.$$

Lemma 4.2 (Weighted switching differential equation)**.** *For s on the imaginary axis,*

$$Z_n'(s) = H_n(s)Z_n(s) + E_n(s), \qquad H_n(s) = \sum_{b_n < r \le U_n} a_{n,r} w_r e^{sw_r}, \tag{28}$$

where

$$\sup_{s \in i\mathbb{R}} |E_n(s)| = o((\log n)^{3/2}). \tag{29}$$

Proof. Apply the marked identity (2) with $\Phi(\sigma) = e^{sB_n^\circ(\sigma)}$. If the source cycle has length $L > r + U_n$, then $L > U_n$ and $L - r > U_n$, whereas $b_n < r \le U_n$. Hence splitting the source cycle changes B_n° by exactly w_r. On the exceptional event $r < L \le r + U_n$, both exponentials have modulus one for imaginary s. It follows that

$$\begin{aligned}&\Big|\mathbb{E}[C_{n,r}e^{sB_n^\circ}] - a_{n,r}e^{sw_r}Z_n(s)\Big| \\ &\quad\le a_{n,r}\Delta_{n,r} + 2a_{n,r}\frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\mathbf{1}_{\{L_n(i)\le r+U_n\}}.\end{aligned} \tag{30}$$

By (13), $r + U_n \le 2U_n \le n/2$. Hölder's inequality, (4), and (8) therefore give

$$\frac{1}{n}\mathbb{E}\sum_{i:L_n(i)>r} R_{i,\sigma_n^r(i)}\mathbf{1}_{\{L_n(i)\le r+U_n\}} \le C_\kappa\left(\frac{U_n}{n}\right)^\beta.$$

Multiply (30) by w_r and sum. On a dyadic block $a_j \le r < 2a_j$, equation (13) gives $a_{n,r} \le 2/r$ for all large n, so

$$\sum_{a_j \le r < 2a_j} w_r a_{n,r}\Delta_{n,r} \le C\log(2a_j)\sup_{a_j \le r < 2a_j}\Delta_{n,r}.$$

Corollary 3.4 handles the sum over blocks. Finally, by (11),

$$\left(\frac{U_n}{n}\right)^\beta \sum_{r \le U_n}\frac{\log r}{r} = O((\log n)^{2 - A_0\beta}) = o((\log n)^{3/2}).$$

This proves (29). □

Lemma 4.3 (Deterministic exponent sums)**.** *With*

$$a_{n,r} = \frac{n}{r(n-r)}, \qquad b_n < r \le U_n,$$

one has

$$\mu_n := \sum_{b_n < r \le U_n} a_{n,r} \log r = \frac{1}{2}(\log n)^2 + O(\log n \log\log n), \tag{31}$$

$$v_n := \sum_{b_n < r \le U_n} a_{n,r} (\log r)^2 = \frac{1}{3}(\log n)^3 + O((\log n)^2 \log\log n), \tag{32}$$

$$\sum_{b_n < r \le U_n} a_{n,r} (\log r)^3 = O((\log n)^4). \tag{33}$$

Proof. Use the exact decomposition

$$a_{n,r} = \frac{n}{r(n-r)} = \frac{1}{r} + \frac{1}{n-r}. \tag{34}$$

The $1/r$ terms are estimated by the corresponding logarithmic integrals. The lower cutoff $b_n = (\log n)^3$ and upper cutoff $U_n = n/(\log n)^{A_0} + O(1)$ produce the displayed logarithmic errors. For $k = 1, 2, 3$, the $1/(n-r)$ contribution is

$$O\left(\frac{U_n}{n}(\log n)^k\right),$$

which is smaller than the stated remainder. This proves (31)–(33). □

Theorem 4.4 (Additive Erdős–Turán law)**.** *Let*

$$\varsigma_n^2 = \frac{1}{3}(\log n)^3.$$

Then

$$\frac{B_n - \frac{1}{2}(\log n)^2}{\varsigma_n} \Rightarrow N(0,1). \tag{35}$$

Proof. Put

$$\Psi_n(s) = \sum_{b_n < r \le U_n} a_{n,r}(e^{s w_r} - 1),$$

so that $\Psi_n' = H_n$. Variation of constants in (28) gives

$$Z_n(s) = e^{\Psi_n(s)} \left[1 + \int_0^s e^{-\Psi_n(z)} E_n(z)\, dz\right]. \tag{36}$$

For $s = it/\varsigma_n$ and $z = iu$ on the integration segment,

$$\begin{aligned} -\Re\Psi_n(iu) &= \sum_{b_n < r \le U_n} a_{n,r}(1 - \cos(u w_r)) \\ &\le \frac{u^2}{2} \sum_{b_n < r \le U_n} a_{n,r} w_r^2. \end{aligned}$$

Lemma 4.3 makes the last sum $O((\log n)^3)$, so $|e^{-\Psi_n(z)}| \le e^{C_t}$ uniformly on the segment. Its length is $O(\varsigma_n^{-1})$, and (29) therefore makes the integral in (36) equal to $o(1)$.

Taylor expansion, together with Lemma 4.3, gives for fixed real t

$$\Psi_n(it/\varsigma_n) - \frac{it}{\varsigma_n}\frac{1}{2}(\log n)^2 = \frac{it}{\varsigma_n}\left(\mu_n - \frac{1}{2}(\log n)^2\right) - \frac{t^2 v_n}{2\varsigma_n^2} + o(1)$$
$$\longrightarrow -\frac{1}{2}t^2.$$

The cubic remainder is bounded by

$$\frac{C_t}{\varsigma_n^3}\sum_{b_n < r \le U_n} a_{n,r}(\log r)^3 = O((\log n)^{-1/2}).$$

Equation (36) and Lévy's theorem prove (35) with B_n° in place of B_n. Lemma 4.1 and Slutsky's theorem restore B_n. □

5 A rate-oriented three-clock switch

Let

$$p(x) = \prod_{a=1}^{n} \theta_a e^{-\theta_a x_a} \mathbf{1}_{\{x_a > 0 \ \forall a\}}$$

be the joint clock density. For a coordinate permutation τ, define $(T_\tau x)_a = x_{\tau(a)}$ and

$$R_\tau(x) = \frac{p(T_\tau x)}{p(x)} = \exp\left\{\sum_{a=1}^{n} \theta_a (x_a - x_{\tau(a)})\right\}. \tag{37}$$

Then

$$\sigma_{T_\tau X} = \sigma_X \circ \tau, \tag{38}$$

and, by the substitution $y = T_\tau x$,

$$\mathbb{E}[R_\tau(X) G(T_\tau X)] = \mathbb{E}G(X) \tag{39}$$

for every nonnegative measurable G.

Fix three distinct labels. Order them deterministically by decreasing rate, with label order used to break ties, and write the resulting order as h_0, h_1, h_2. Define

$$\tau^* = (h_0\, h_1\, h_2).$$

This is exactly one of the two cyclic orientations of the three labels. Moreover,

$$\log R_{\tau^*} = (\theta_{h_0} - \theta_{h_2})X_{h_0} - (\theta_{h_0} - \theta_{h_1})X_{h_1} - (\theta_{h_1} - \theta_{h_2})X_{h_2}. \tag{40}$$

Thus only the maximum-rate clock has a positive coefficient.

For $r, s \ge 1$ with $r + s < n$ and a source i satisfying $L_n(i) > r + s$, define

$$j_i^{(1)} = \sigma_n^r(i), \qquad k_i^{(1)} = \sigma_n^{r+s}(i), \qquad \tau_i^{(1)} = (i\, k_i^{(1)}\, j_i^{(1)}),$$
$$k_i^{(2)} = \sigma_n^s(i), \qquad j_i^{(2)} = \sigma_n^{r+s}(i), \qquad \tau_i^{(2)} = (i\, j_i^{(2)}\, k_i^{(2)}).$$

Set

$$\mathcal{I}_{n;r,s} = \sum_{i:L_n(i)>r+s} \Big[\mathbf{1}_{\{\tau^*(i,j_i^{(1)},k_i^{(1)})=\tau_i^{(1)}\}} R_{\tau_i^{(1)}}(X) + \mathbf{1}_{\{\tau^*(i,j_i^{(2)},k_i^{(2)})=\tau_i^{(2)}\}} R_{\tau_i^{(2)}}(X) \Big]. \tag{41}$$

Here $\tau^*(i,j,k)$ denotes the rate-oriented cycle on the set $\{i,j,k\}$. When $r=s$, the two displayed terms may coincide as random variables; they are retained with multiplicity because the target marks order the two r-cycles.

Proposition 5.1 (Exact three-clock factorial identity)**.** *For $r,s \geq 1$ with $r+s<n$,*

$$rs(n-r-s)\mathbb{E}\big[C_{n,r}(C_{n,s}-\mathbf{1}_{\{r=s\}})\big] = \mathbb{E}\mathcal{I}_{n;r,s}. \tag{42}$$

Pointwise, $\mathcal{I}_{n;r,s}$ is a sum of at most $2n$ nonnegative three-clock factors, counted with multiplicity.

Proof. For a permutation π, let $A_{r,s}(\pi;i,j,k)$ be the event that j lies in an r-cycle of π, k lies in a distinct s-cycle, and i lies in neither cycle. The exact number of target marks is

$$\sum_{i,j,k \text{ distinct}} \mathbf{1}_{A_{r,s}(\pi;i,j,k)} = rs(n-r-s)C_r(\pi)(C_s(\pi)-\mathbf{1}_{\{r=s\}}). \tag{43}$$

For every deterministic triple (i,j,k), equation (39) gives

$$\mathbb{E}\mathbf{1}_{A_{r,s}(\sigma_X;i,j,k)} = \mathbb{E}\left[R_{\tau^*(i,j,k)}(X)\mathbf{1}_{A_{r,s}(\sigma_X\circ\tau^*(i,j,k);i,j,k)}\right]. \tag{44}$$

If $\tau^*(i,j,k) = (i\,k\,j)$, then

$$A_{r,s}(\sigma\circ(i\,k\,j);i,j,k) \iff L_\sigma(i)>r+s, \quad j=\sigma^r(i), \quad k=\sigma^{r+s}(i). \tag{45}$$

Indeed, the source cycle is traversed from i to j in r steps, from j to k in s steps, and then back to i. Right-composition by $(i\,k\,j)$ replaces the three outgoing edges so as to cut off an r-cycle containing j, an s-cycle containing k, and a residual cycle containing i. The operation is reversible.

Similarly, if $\tau^*(i,j,k) = (i\,j\,k)$, then

$$A_{r,s}(\sigma\circ(i\,j\,k);i,j,k) \iff L_\sigma(i)>r+s, \quad k=\sigma^s(i), \quad j=\sigma^{r+s}(i). \tag{46}$$

Here the source cycle is traversed from i to k in s steps and from k to j in r steps. Again the right-composition cuts the source cycle into the required three marked cycles and is reversible.

Sum (44) over all distinct triples, use (43), and reorganize the right side by the source label i using (45) and (46). The result is exactly (41), proving (42). There are at most two terms for each source i. □

Lemma 5.2 (Selected three-clock tail bound)**.** *For $\kappa>1$ and $u\geq 0$,*

$$\frac{1}{n}\mathbb{E}\sum_{\text{summands in } \mathcal{I}_{n;r,s}} \mathbf{1}_{\{\log R_{\tau^*}>u\}} \leq 6e^{-\lambda_\kappa u}, \tag{47}$$

uniformly in n,r,s. Consequently,

$$\mathbb{E}\mathcal{I}_{n;r,s} \leq C_\kappa n. \tag{48}$$

For $\kappa=1$, every factor equals one and the same conclusion holds.

Proof. Consider a selected summand and let h be its maximum-rate label. If all three rates are equal, then $R_{\tau^*} = 1$, so there is nothing to prove for $u > 0$. Otherwise, equation (40) implies

$$\{\log R_{\tau^*} > u\} \subseteq \left\{ X_h > \frac{u}{\theta_h - \theta_{\min}} \right\} \subseteq \left\{ X_h > \frac{u}{\theta_h - 1} \right\}.$$

Since $X_h \sim \mathrm{Exp}(\theta_h)$ and $x/(x-1)$ is decreasing on $(1, \kappa]$,

$$\mathbb{P}\left\{ X_h > \frac{u}{\theta_h - 1} \right\} = \exp\left\{ -\frac{\theta_h}{\theta_h - 1} u \right\} \leq e^{-\lambda_\kappa u}. \tag{49}$$

Partition the selected summands according to the two orientation types in (41) and according to whether h is the source, the first endpoint, or the second endpoint. There are at most six classes. Within each class, the map from the selected source i to h is the restriction of one of

$$i \mapsto i, \qquad i \mapsto \sigma_n^r(i), \qquad i \mapsto \sigma_n^s(i), \qquad i \mapsto \sigma_n^{r+s}(i).$$

Within a fixed class, the position of the maximum-rate label relative to the source is fixed, so the map has the form $i \mapsto \sigma_n^t(i)$ for one fixed $t \in \{0, r, s, r+s\}$. Since σ_n^t is a permutation of $[n]$, its restriction to the selected source set is injective. Reindexing and (49) therefore give at most $ne^{-\lambda_\kappa u}$ per class, proving (47).

Let N_u be the number of selected summands with $\log R_{\tau^*} > u$. Since there are at most $2n$ summands,

$$\mathcal{I}_{n;r,s} \leq 2n + \int_0^\infty e^u N_u \, du.$$

Taking expectations and applying (47),

$$\mathbb{E}\mathcal{I}_{n;r,s} \leq 2n + 6n \int_0^\infty e^{-(\lambda_\kappa - 1)u} \, du \leq C_\kappa n.$$

If $\kappa = 1$, every summand is one and $\mathcal{I}_{n;r,s} \leq 2n$ directly. □

Corollary 5.3 (Uniform two-cycle factorial bound)**.** *If $r + s \leq n/2$, then*

$$\mathbb{E}\big[C_{n,r}(C_{n,s} - \mathbf{1}_{\{r=s\}})\big] \leq \frac{C_\kappa}{rs}. \tag{50}$$

Proof. Combine (42) and (48), using $n - r - s \geq n/2$. □

6 The product–least-common-multiple comparison

Let

$$\mathcal{P}_n = \{r : b_n < r \leq U_n,\ C_{n,r} > 0\}, \qquad O_n^\circ = \operatorname{lcm} \mathcal{P}_n,$$

with the convention $\operatorname{lcm} \varnothing = 1$. Let Λ be the von Mangoldt function,

$$\Lambda(m) = \begin{cases} \log p, & m = p^a \text{ for a prime } p \text{ and } a \geq 1, \\ 0, & \text{otherwise,} \end{cases}$$

and define, for $m \geq 1$,

$$D_{n,m} = \sum_{\substack{b_n < r \leq U_n \\ m \mid r}} C_{n,r}.$$

The identities

$$\log r = \sum_{m \mid r} \Lambda(m)$$

and

$$\log O_n^\circ = \sum_{m \le U_n} \Lambda(m) \mathbf{1}_{\{D_{n,m} \ge 1\}}$$

give the exact decomposition

$$B_n^\circ - \log O_n^\circ = \sum_{m \le U_n} \Lambda(m)(D_{n,m} - 1)_+. \tag{51}$$

Proposition 6.1 (Arithmetic overlap bound)**.** *One has*

$$\mathbb{E}(B_n^\circ - \log O_n^\circ) = O_\kappa(\log n \log\log n). \tag{52}$$

Proof. Put $y_n = (\log n)^2$. For $m \le y_n$, use $(D_{n,m} - 1)_+ \le D_{n,m}$. Since $U_n \le n/2$ for all large n, Lemma 2.2 gives

$$\mathbb{E} D_{n,m} \le \frac{C_\kappa}{m} \sum_{j \le U_n/m} \frac{1}{j} \le C_\kappa \frac{\log n}{m}.$$

Chebyshev's elementary estimate [7]

$$\psi(x) := \sum_{m \le x} \Lambda(m) = O(x)$$

and partial summation imply

$$\sum_{m \le y} \frac{\Lambda(m)}{m} = O(\log y). \tag{53}$$

Thus the contribution of $m \le y_n$ is $O(\log n \log\log n)$.

For a nonnegative integer D, $(D-1)_+ \le (D)_2/2$. Moreover,

$$(D_{n,m})_2 = \sum_{\substack{b_n < r,s \le U_n \\ m \mid r,\ m \mid s}} C_{n,r}(C_{n,s} - \mathbf{1}_{\{r=s\}}). \tag{54}$$

By (13), every contributing pair satisfies $r + s \le 2U_n \le n/2$. Corollary 5.3 therefore gives

$$\begin{aligned} \mathbb{E}(D_{n,m})_2 &\le C_\kappa \left(\sum_{j \le U_n/m} \frac{1}{mj} \right)^2 \\ &\le C_\kappa \frac{\log^2(1 + U_n/m)}{m^2}. \end{aligned} \tag{55}$$

A second partial summation using $\psi(x) = O(x)$ gives

$$\sum_{m > y} \frac{\Lambda(m)}{m^2} = O(1/y). \tag{56}$$

Consequently, the contribution of $m > y_n$ is at most

$$C_\kappa (\log n)^2 \sum_{m > y_n} \frac{\Lambda(m)}{m^2} = O_\kappa(1).$$

Combining the two ranges proves (52). □

Let

$$O_n = \mathrm{ord}(\sigma_n).$$

Corollary 6.2 (Product–LCM discrepancy)**.** *One has*

$$\mathbb{E}[B_n - \log O_n] = O_\kappa(\log n \log\log n). \tag{57}$$

In particular,

$$B_n - \log O_n = o_{L^1}((\log n)^{3/2}). \tag{58}$$

Proof. The least common multiple O_n° divides O_n, while $\log O_n \le B_n$. Hence the deterministic inequality

$$0 \le B_n - \log O_n \le (B_n - B_n^\circ) + (B_n^\circ - \log O_n^\circ) \tag{59}$$

holds. Apply Lemma 4.1 and Proposition 6.1. □

7 Proof of the main theorem

Proof of Theorem 1.1. After a common rescaling, the hypothesis of Theorem 1.1 is exactly (1). By Corollary 6.2,

$$\frac{\log \mathrm{ord}(\sigma_n) - B_n}{\sqrt{\frac{1}{3}(\log n)^3}} \longrightarrow 0$$

in L^1, hence in probability. Theorem 4.4 and Slutsky's theorem complete the proof. □

Remark 7.1 (Quantitative refinement). The proof deliberately retains the explicit endpoint error, the growing truncation dependence, and the L^1 product–LCM discrepancy. These estimates form the input to a separate Berry–Esseen analysis. No rate of normal approximation is asserted in the present paper.

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